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Guided Reasoning Chains for API Recommendation

Kenji Matsumoto and Yohei Murakami
Graduate School of Information Science and Engineering
Ritsumeikan University, Japan

Research Objective

- Rapid growth of available APIs makes it hard for users to discover and compose APIs that satisfy their requirements

Related work

- APIs with multi-label classification models trained on vectorized natural-language requirements and API features^[1]
 - The reasons for recommendations have low explainability

Proposed method

LLM-based method for stable and explainable API recommendation



Requirements



LLM



Google Maps :for delivery

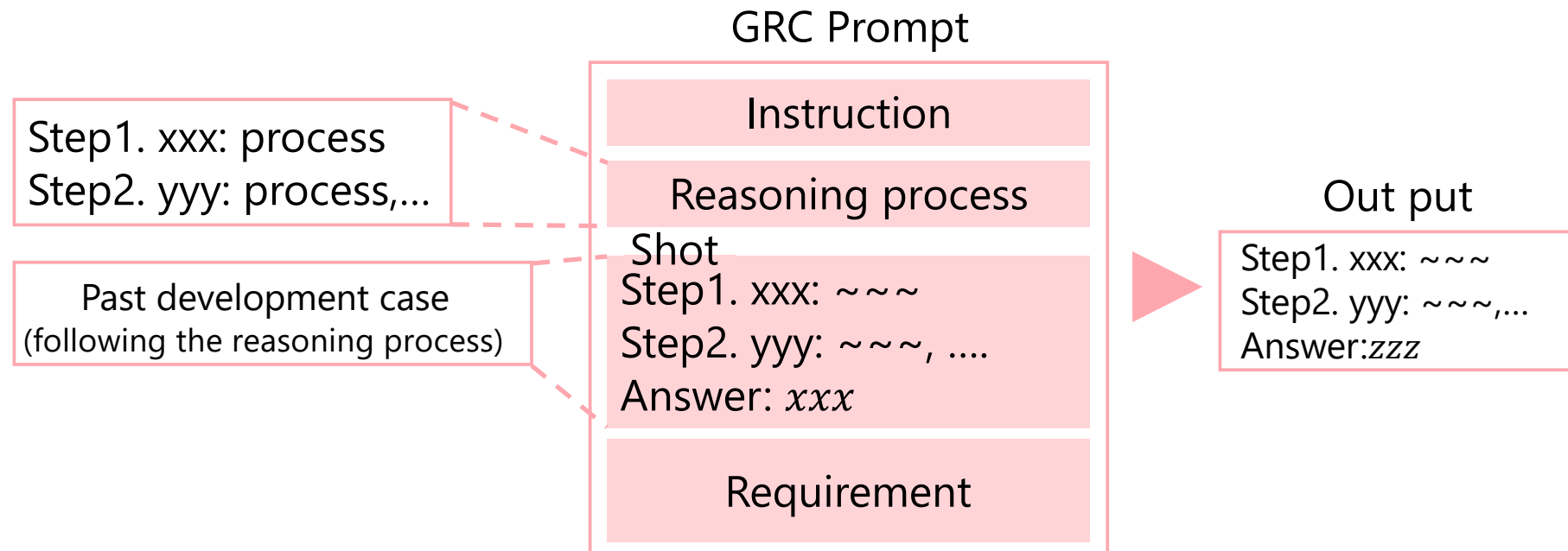


stripe : for payment
Recommendation

[1] Wu, H., Duan, Y., Yue, K., Zhang, L.: Mashup-oriented Web API Recommendation via Multi-Model Fusion and Multi-Task Learning. IEEE Trans. Serv. Compute. **15**(6), 3330–3343 (2021)

Guided Reasoning Chains (GRC)

- Traditional prompting methods only provides examples, so the model's reasoning is free-format and often inconsistent or hard to explain
- GRC is a prompting framework for controlling how an LLM reasons
 - Predefines a task-specific **reasoning process**
 - Embeds both the reasoning process and example cases(shot) that follow this reasoning process into the prompt



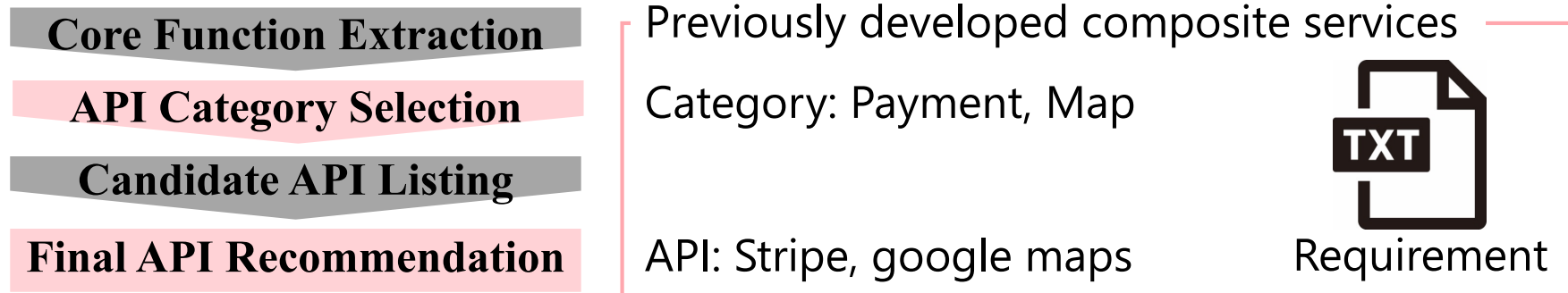
Reasoning process in Service Composition

- The reasoning process is a top-down functional decomposition in software engineering from a requirement to concrete APIs
 - Requirements are often written without explicit functions
 - extract core functions
 - Many APIs make direct mapping from core functions difficult
 - narrow the search space at the API-category level, then select APIs
- Requirement: A service to easily **order** food and have it **delivered** quickly

Core Function Extraction	Ordering	Delivery
API Category Selection	Payment	Map
Candidate API Listing	 PayPal stripe	 Bing maps Google Maps
Final API Recommendation	stripe	Google Maps

Shot Generation for API Recommendation

- Shots perform interpolative reasoning along the reasoning process
 - They Interpolate from the requirement and listed API categories/APIs



Shot generation Prompt

Core Function Extraction	Ordering	Delivery
API Category Selection	Payment	Map
Candidate API Listing	PayPal, Stripe	Bing Maps, Google Maps
Final API Recommendation	Stripe	Google Maps

Shot Generation for API Recommendation

Prompt

Instruction

This is a system for creating reasoning examples that propose appropriate APIs based on the requirements of a mashup service, following the given reasoning process.

Rule

Step 1: extract core functions from the Requirements that can lead to the given Correct categories.

Step 2: select only the Correct categories.

Step 3: from the available APIs, list several plausible candidate APIs, must including the Correct APIs

Step 4: select only Correct APIs.

#Insert Reasoning process

API Information

Available categories: [category, ...]

Available APIs: {API_name : [category, description], ...}

#Insert Requirement

API usage information in requirements

Correct categories: [Category names used in the requirement]

Correct APIs: [API names used in the requirement]

#Insert Output Format

Overview of the API recommendation prompt

- Uses the reasoning process and shots to recommend API combinations from a requirement

Prompt

Instruction

This is a system for proposing appropriate APIs based on the requirements of a mashup service.

#Insert Reasoning process

#Insert Shot

API Information

Available categories: [category, ...]

Available APIs: {API_name : [category, description], ...}

#Insert Requirement

#Insert Output Format

Details of the API recommendation

Instruction

Instruction to recommend APIs that match the requirement

Reasoning process

Steps to recommend APIs

Shot

Examples of mashup services

- Created based on the **Reasoning process**

API Information

API name, Category list, API description

Requirement

Text for the mashup service to be created

Output Format

Clearly specify the answer format

Baselines

- All methods use the same Instruction prompt as the proposed method

Baseline Methods

- **Zero Shot**

- Only the requirement

- **Zero Shot CoT**

- Add the phrase "Let's think step by step"

- **Few Shot**

- Each shot is a pair of a requirement and its category/API

- **Few Shot CoT**

- Each shot includes reasoning, using GPT-4.1-mini outputs
- The reasoning is free-form and does not use a structured reasoning process

- **Plan and Solve**

- Add the phrase "Let's first understand the problem and devise a plan to solve it. Then, let's carry out the plan and solve the problem step by step"

Data Used

- Service information collected from ProgrammableWeb
 - Mashup service: 6,184, API:2,746
- Filtering Conditions
 - Mashup service: Use two or more APIs $\cap$ Complete API information
 - API: Only API used in mashup service
- Evaluation Data
 - Mashup service: 1,497, API data:645

Mashup Service Information

name, used API name, description

API Information

name, category, description

Evaluation Method

- Evaluate categories and final APIs within the Reasoning process
 - requirement: description of the mashup service
 - Correct data: created from API name and category used in the requirement
- number of shots used in the prompt: 6 (random)
- LLMs used
 - GPT-4.1 mini
 - Llama3.3:70b
 - Deepseek-R1:70B
- Evaluation metric
 - F1-score

Evaluation

	Category / API (F1-score)		
	GPT	LLAMA	DeepSeek
Zero Shot	0.661/0.376	0.313/0.082	0.326/0.065
Zero Shot CoT	0.644/0.367	0.319/0.066	0.340/0.062
Few Shot	0.669/0.444	0.301/0.065	0.194/0.064
Few Shot CoT	0.653/0.420	0.160/0.057	0.235/0.057
Plan and Solve	0.620/0.361	0.322/0.084	0.315/0.071
Ours	0.683/0.450	0.339/0.095	0.342/0.084

- The proposed method achieves higher accuracy in both inferences

Ablation study

- We conduct an ablation study to verify proposed method
 - Use the same Instruction prompt as the proposed method

Ablation Methods

- **Zero Shot Infer** (Tests effect of shots)
 - Using only the Requirement and the reasoning process
- **Few Shot ablation** (Tests effect of reasoning process prompt)
 - Removing the reasoning process prompt from the proposed method

	Category / API (F1-score)		
	GPT	LLAMA	DeepSeek
Zero Shot Infer	0.621/0.366	0.335/0.52	0.341/0.070
Few Shot ablation	0.682/0.421	0.324/0.063	0.330/0.081
Ours	0.683/0.450	0.339/0.095	0.342/0.084

Accuracy Change by Shot Selection

- In few-shot learning, the choice of shots greatly affects accuracy
 - Accuracy is evaluated by similarity between the requirement and the shot
- Similarity is calculated using semantic embedding model¹
- Compared with the following four methods (number of shots: 6)
 - Random
 - Nearest : most similar shots
 - Furthest: least similar shots
 - Nearest-Furthest: 3 most similar shots+ 3 least similar shots
- Model : GPT-4.1 mini

¹ <https://huggingface.co/intfloat/e5-large-v2>

Accuracy Change by Shot Selection

Category / API(F1-score)	
	GPT
Random	0.683/0.450
Nearest	0.688/0.557
Furthest	0.639/0.417
Nearest - Farthest	0.678/0.511

- Category recommendation(F1)
 - Nearest > Random > Nearest-Furthest > Furthest
- API recommendation(F1)
 - Nearest > Nearest-Furthest > Random > Furthest
- Selecting shots that semantically match the input requirement is important

Hallucination Analysis

- hallucination: number of requirements with at least one category or API not in the given API information.

Requirements with hallucinated categories / APIs

N=1497 requirements

	Category	API
Zero Shot CoT	47	181
Few Shot CoT	9	59
Plan and solve	6	141
Ours	1	55

- The proposed method shows better performance not only in accuracy but also in output reliability

Conclusion

- We proposed GRC as a prompting framework for LLM-based API recommendation.
- The reasoning process is a top-down functional decomposition in software engineering, from the requirement to concrete APIs
- This reasoning process and the shots improve F1-scores for both categories and APIs over baselines
- GRC also reduces hallucinated categories/APIs, so the recommended APIs are more reliable and explainable

Future work

- Extend the reasoning process based on GORE (Goal-Oriented Requirements Engineering)
 - We will decompose each function recursively to better match requirements with APIs

What is Guided Reasoning Chains(GRC)

- A framework that guides LLMs to follow task-specific structured reasoning process

Few Shot CoT

Input

Q. He has 5 balls. He buys 2 cans, and each can have 3 balls. How many balls does he have now?

A. He already has 5 balls. 2 cans have 3 balls each, so that is 6 more balls. The answer is 11.

Q. Tom has 7 notebooks. He buys 3 packs of notebooks, and each pack has 4 notebooks. How many notebooks does he have now?

Output

A. First, Tom has 7 notebooks. He then buys 3 packs, and pack has 4 notebooks, so there are $3 + 4 = 7$ new notebooks. Now he has 7 original notebooks and 7 new notebooks. So, he has 14 notebooks in total. **✗**

Proposed method Reasoning Process

Input

Q. He has 5 balls. He buys 2 cans, and each can have 3 balls. How many balls does he have now?

A.

1. Initial number: He has 5 balls at first.
2. Additional Purchase: He buys 2 cans. Each can have 3 balls. $2 \times 3 = 6$
3. Total Calculation: $5 + 6 = 11$

Q. ##### same Few Shot CoT #####

Output

A.

1. Initial number: He has 7 notebooks at first.
2. Additional purchase: He buys 3 packs, and each pack has 4 notebooks. $3 \times 4 = 12$
3. Total Calculation: $7 + 12 = 19$ ○

} Shot

Reasoning process prompt

Core Function Extraction

API Category Selection

Candidate API Listing

Final API Recommend

Prompt

It follows the reasoning process below:

1. Infers the core functionalities of the mashup service based on the given Requirements.
2. Identify and propose relevant API categories from the Available categories.
3. Match APIs from each category based on descriptions and requirement alignment.
4. Select and recommend the best-suited APIs with reasons.

Output example of shot generation prompt

- Use requirement/output as **shot** for the API recommendation prompt

requirement: Breakout is the first modern Workflow Management Software that automates recurring business processes.

Output

1. Core Function Extraction

Automation of Business Processes : reason,
Data Handling and Documentation: reason, ...

2. API Category Selection :

Email : reason, Spreadsheets : reason,
Storage : reason, Tools : reason

3. Candidate API Listing

Email :
Mail chimp : reason, Gmail: reason, Mail jet: reason, ...
Spreadsheets : ...

4. Final API Recommendation

Gmail, Google-sheets, Google-drive, Slack-web

Rule

- Step2: select only requirement categories
- Step3: include requirement APIs
- Step4: select only requirement APIs

Ablation prompt

Proposed method

Instruction

Reasoning process

Shot

API Information

Requirement

Output Format

Zero Shot Infer

Instruction

Reasoning process

Shot

API Information

Requirement

Output Format

Few Shot ablation

Instruction

Reasoning process

Shot

API Information

Requirement

Output Format



Output format
